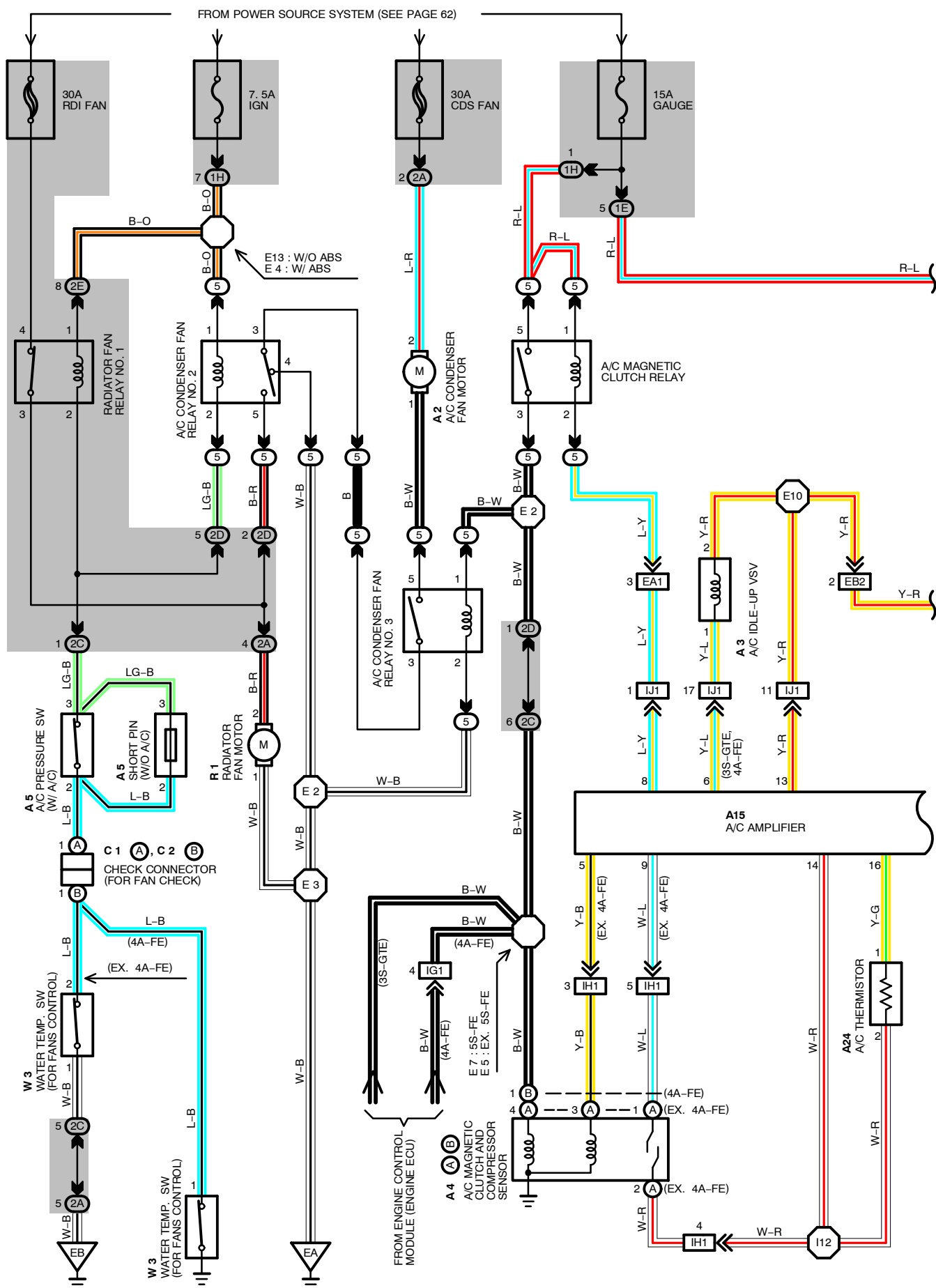
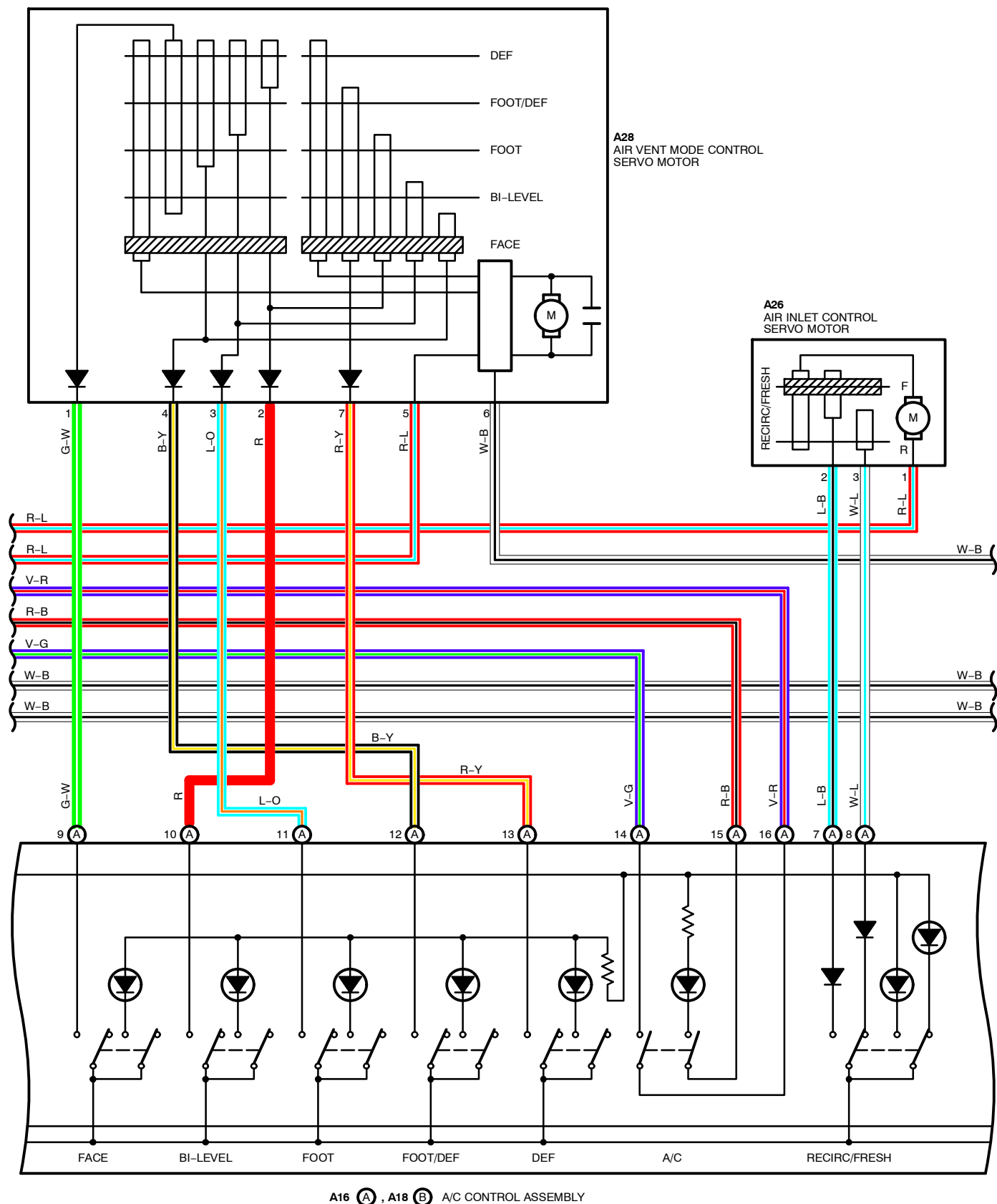


RADIATOR FAN AND AIR CONDITIONING (MANUAL AIR CONDITIONING)





RADIATOR FAN AND AIR CONDITIONING (MANUAL AIR CONDITIONING)



RADIATOR FAN AND AIR CONDITIONING (MANUAL AIR CONDITIONING)

SYSTEM OUTLINE

1. COOLING FAN OPERATION

WHEN THE IGNITION SW IS TURNED ON, THE CURRENT FROM IGN FUSE FLOWS TO **TERMINAL 1** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 2** → **TERMINAL 3** OF THE A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 1** (4S-FE) OR **TERMINAL 2** (EX. 4A-FE) OF THE WATER TEMP. SW → **TERMINAL 1** (EX. 4A-FE) → **GROUND**, FROM **TERMINAL 1** OF A/C FAN RELAY NO. 2 → **TERMINAL 2** → **TERMINAL 3** OF A/C PRESSURE SW → **TERMINAL 2** → **TERMINAL 1** (4A-FE) OR **TERMINAL 2** (EX. 4A-FE) OF THE WATER TEMP. SW → **TERMINAL 1** (EX. 4A-FE) → **GROUND**, CAUSING RELAY NO. 1 AND RELAY NO. 2 OF EACH FAN TO TURN ON.

* OPERATION AT LOW SPEED

WHEN THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON AND THE A/C OPERATES, THE CURRENT FLOWS FROM GAUGE FUSE FLOWS TO **TERMINAL 1** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** → **TERMINAL 8** OF A/C AMPLIFIER CAUSING A/C MAGNETIC CLUTCH RELAY TO TURN ON.

AT THAT TIME, THE CURRENT FROM GAUGE FUSE FLOWS TO **TERMINAL 5** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 3** → **TERMINAL 4** (5S-FE), 1 (4A-FE) OF A/C MAGNETIC CLUTCH → **GROUND**, AND FROM **TERMINAL 3** OF A/C MAGNETIC CLUTCH RELAY → **TERMINAL 2** OF A/C FAN RELAY NO. 3 → **TERMINAL 2** → **GROUND**.

AS A RESULT, A/C MAGNETIC CLUTCH AND A/C FAN RELAY NO. 3 TURNS ON AND THE CURRENT FLOWS FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 5** → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **GROUND**, FLOWING TO EACH FAN MOTOR IN SERIES, CAUSING THE COOLING FAN TO ROTATE AT LOW SPEED.

* OPERATION AT HIGH SPEED

DURING A/C OPERATION, WHEN THE PRESSURE OF A/C COMPRESSOR BECOMES HIGHER THAN NORMAL PRESSURE (MORE THAN 14.3 KG/CM² 1401 KPA, 203 PSI), THE A/C PRESSURE SW TURNS OFF.

AS A RESULT, FAN RELAY NO. 1 AND NO. 2 TURNS OFF AND THE CURRENT FLOWS FROM RDI FAN FUSE → **TERMINAL 4** OF RADIATOR FAN RELAY NO. 1 → **TERMINAL 3** → **TERMINAL 2** OF RADIATOR FAN MOTOR → **TERMINAL 1** → **GROUND**, AND FROM CDS FAN FUSE → **TERMINAL 2** OF A/C CONDENSER FAN MOTOR → **TERMINAL 1** → **TERMINAL 5** OF A/C FAN RELAY NO. 3 → **TERMINAL 3** → **TERMINAL 3** OF A/C FAN RELAY NO. 2 → **TERMINAL 4** → **GROUND**, FLOWING TO EACH FAN MOTOR IN PARALLEL CAUSING THE COOLING FAN TO ROTATE AT HIGH SPEED. WHEN THE ENGINE COOLANT TEMPERATURE BECOMES MORE THAN ABOUT 90°C (194°F), THE WATER TEMP. SW TURNS OFF AND THE SAME OPERATION AS ABOVE IS PERFORMED.

2. HEATER BLOWER MOTOR OPERATION (DIAL TYPE BLOWER (W/O AUTO A/C))

CURRENT IS APPLIED AT ALL TIMES THROUGH THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY. WHEN THE IGNITION SW IS TURNED ON, CURRENT FLOWS THROUGH GAUGE FUSE TO **TERMINAL 3** OF HEATER RELAY → **TERMINAL 1** → **TERMINAL (B) 2** AND **TERMINAL (B) 7** OF A/C CONTROL ASSEMBLY.

* LOW SPEED OPERATION

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **LO** POSITION. THE CURRENT FLOWS TO **TERMINAL (B) 7** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF THE HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 4** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT LOW SPEED.

* HIGH SPEED OPERATION

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **HI** POSITION. THE CURRENT TO **TERMINAL (B) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, THE CURRENT TO **TERMINAL 5** OF HEATER RELAY → **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL (B) 5** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**, AND CAUSES THE BLOWER MOTOR TO ROTATE AT HIGH SPEED.

* MEDIUM SPEED OPERATION (OPERATION AT MANUAL M1, M2)

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **M1** POSITION. THE CURRENT FLOWING TO **TERMINAL 3** OF HEATER RELAY FLOWS TO **TERMINAL 1** → **TERMINAL (B) 2** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND** AND TURNS THE HEATER RELAY ON.

AS A RESULT, CURRENT FROM THE HEATER FUSE TO **TERMINAL 5** OF HEATER RELAY FLOW TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 3** → **TERMINAL (B) 4** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**, CAUSING THE BLOWER MOTOR TO ROTATE AT MEDIUM LOW SPEED.

WHEN THE BLOWER SW (A/C CONTROL ASSEMBLY) IS MOVED TO **M2** POSITION. THE CURRENT TO **TERMINAL 5** OF HEATER RELAY FLOWS TO **TERMINAL 4** → **TERMINAL 2** OF BLOWER MOTOR → **TERMINAL 1** → **TERMINAL 1** OF BLOWER RESISTOR → **TERMINAL 2** → **TERMINAL (B) 6** OF A/C CONTROL ASSEMBLY → **TERMINAL (B) 3** → **GROUND**.

AS THE CURRENT FLOW FROM BLOWER MOTOR TO GROUND IS GREATER THAN FOR M1. THE BLOWER MOTOR ROTATES AT HIGH SPEED.

3. OPERATION OF AIR INLET CONTROL SERVO MOTOR

(SWITCHING FROM FRESH TO RECIRC)

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR. WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE RECIRC SIDE, TO CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 2** → **TERMINAL (A) 7** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE RECIRC SIDE.

WHEN IT IS IN THE **RECIRC** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

(SWITCHING FROM RECIRC TO FRESH)

WITH IGNITION SW ON, WHEN THE RECIRC/FRESH SW IS SWITCHED TO THE FRESH SIDE, THE CURRENT FLOWS FROM **TERMINAL 1** OF AIR INLET CONTROL SERVO MOTOR → **TERMINAL 3** → **TERMINAL (A) 8** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**, THE MOTOR ROTATES AND THE DAMPER MOVES TO THE FRESH SIDE. WHEN IT IS IN THE **FRESH** POSITION, THE CURRENT IS CUT INSIDE THE SERVO MOTOR AND THE DAMPER STOPS AT THAT POSITION.

4. OPERATION OF AIR VENT MODE CONTROL SERVO MOTOR

WITH IGNITION SW TURNED ON, THE CURRENT FLOWS FROM GAUGE FUSE TO **TERMINAL 5** OF AIR VENT MODE CONTROL SERVO MOTOR → **TERMINAL 6** → **GROUND**, AND THE DAMPER MOVES TO THE POSITION OF THE MODE SELECTION SW OF THE CONTROL ASSEMBLY SWITCH IS ON.

WHEN THE MODE SELECTION SW OF A/C CONTROL ASSEMBLY IS MOVED TO **DEF** POSITION FROM THE DAMPER IN THE **FACE** POSITION, THE CURRENT FLOWS FROM **TERMINAL 7** OF AIR VENT MODE CONTROL SERVO MOTOR TO **TERMINAL (A) 13** OF A/C CONTROL ASSEMBLY → **TERMINAL (A) 2** → **GROUND**.

AS A RESULT, TO SERVO MOTOR OPERATES UNTIL THE DAMPER REACHES **DEF** POSITION.

WHEN THIS OCCURS THE CURRENT TO THE A/C CONTROL ASSEMBLY IS SHUT OFF AND ROTATION OF THE MOTOR STOPS. SWITCHING TO OTHER MODES IS CONTROLLED BY THE SERVO MOTOR ACCORDING THE FLOWING CURRENT:

1. **FOOT/DEF** POSITION: THE CURRENT FLOWS FROM **TERMINAL 4** OF SERVO MOTOR TO **TERMINAL (A) 12** OF A/C CONTROL ASSEMBLY.
2. **FOOT** POSITION: THE CURRENT FLOWS FROM **TERMINAL 3** OF SERVO MOTOR TO **TERMINAL (A) 11** OF A/C CONTROL ASSEMBLY.
3. **BI-LEVEL** POSITION: THE CURRENT FLOWS FROM **TERMINAL 2** OF SERVO MOTOR TO **TERMINAL (A) 10** OF A/C CONTROL ASSEMBLY.

5. AIR CONDITIONING OPERATION

THE A/C AMPLIFIER RECEIVES VARIOUS SIGNALS, THE ENGINE RPM SIGNAL FROM THE IGNITER, OUTLET TEMPERATURE SIGNAL FROM THE A/C THERMISTOR, COOLANT TEMPERATURE FROM THE ENGINE COOLANT TEMP. SENSOR (EFI WATER TEMP. SENSOR), AND THE LOCK SIGNAL FROM THE A/C COMPRESSOR, ETC.

WHEN THE ENGINE IS STARTED AND THE A/C SW (A/C CONTROL ASSEMBLY) IS TURNED ON, THE CURRENT FLOWS FROM A/C FUSE TO **TERMINAL 13** OF A/C AMPLIFIER.

AS A RESULT, THE CURRENT FLOWING FROM **TERMINAL 2** OF A/C MAGNETIC CLUTCH RELAY TO **TERMINAL 8** OF A/C AMPLIFIER FLOWS FROM **TERMINAL 15** OF A/C AMPLIFIER TO **GROUND** AND TURNS IN THE MAGNETIC CLUTCH RELAY.

BECAUSE THE MAGNETIC CLUTCH IS ON, THE A/C COMPRESSOR OPERATES, CAUSING THE CURRENT FLOWING FROM A/C IDLE-UP VSV TO **TERMINAL 6** OF A/C AMPLIFIER TO FROM TO **TERMINAL 15** OF A/C AMPLIFIER → **GROUND**, AND TURNS ON THE VSV TO AVOID LOWERING THE ENGINE RPM DURING A/C OPERATION.

WHEN ANY OF THE FOLLOWING SIGNALS ARE INPUT TO THE A/C AMPLIFIER, THE AMPLIFIER OPERATES TO TURN OFF THE A/C:

- * ENGINE HIGH RPM SIGNAL
- * COOLANT TEMP. SIGNAL IS HIGH
- * A SIGNAL THAT THE TEMPERATURE AT THE AIR OUTLET IS LOW
- * A SIGNAL THAT THERE IS A LARGE DIFFERENCE BETWEEN ENGINE RPM AND COMPRESSOR RPM
- * A SIGNAL THAT THE REFRIGERANT PRESSURE IS ABNORMALLY HIGH OR LOW

RADIATOR FAN AND AIR CONDITIONING (MANUAL AIR CONDITIONING)

SERVICE HINTS

A 4 (A)(B) A/C MAGNETIC CLUTCH

(A)4, (B)1-GROUND : APPROX. 3.7 Ω

A 5 A/C PRESSURE SW

3-2 : OPEN ABOVE APPROX. 13.5 KG/CM² (192 PSI, 1323 KPA)

CLOSED BELOW APPROX. 10 KG/CM² (142 PSI, 980 KPA)

1-4 : OPEN WITH PRESSURE LESS THAN 2.1 KG/CM² (30 PSI, 206 KPA) OR ABOVE 27 KG/CM² (384 PSI, 2648 KPA)

A15 A/C AMPLIFIER

8-15 : CONTINUITY WITH A/C SW (A/C CONTROL ASSEMBLY) ON AND IGNITION SW AT ON POSITION

14-15 : ALWAYS CONTINUITY

14-GROUND: ALWAYS CONTINUITY

15-GROUND: ALWAYS CONTINUITY

13-GROUND: APPROX. 12 VOLTS WITH IGNITION SW ON

A24 A/C THERMISTOR

1-2 : APPROX. 4852 Ω AT 0°C (32°F)

APPROX. 2341 Ω AT 15°C (39°F)

APPROX. 1500 Ω AT 25°C (77°F)

B 6 BLOWER RESISTOR

1-3 : APPROX. 0.48 Ω

3-2 : APPROX. 0.94 Ω

2-4 : APPROX. 0.91 Ω

W 3 WATER TEMP. SW (FOR FANS CONTROL)

1-2 : OPEN ABOVE APPROX. 90°C (194°F)

CLOSED BELOW APPROX. 83°C (181.4°F)

○ : PARTS LOCATION

CODE	SEE PAGE	CODE	SEE PAGE	CODE	SEE PAGE
A 2	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	A18 A	32	C 2 B	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)
A 3	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	A24	32	J 4	33
A 4	A 28 (3S-GTE), 31 (4A-FE)	A26	32	J 5	33
	B 28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	A28	32	J 6	33
A 5	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)	B 4	32	R 1	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)
A15	32	B 6	32	W 3	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)
A16 A	32	C 1 A	28 (3S-GTE), 30 (5S-FE), 31 (4A-FE)		

○ : RELAY BLOCKS

CODE	SEE PAGE	RELAY BLOCKS (RELAY BLOCK LOCATION)
4	27	R/B NO. 4 (RIGHT KICK PANEL)
5	27	R/B NO. 5 (ENGINE COMPARTMENT FRONT RIGHT)

○ : JUNCTION BLOCK AND WIRE HARNESS CONNECTOR

CODE	SEE PAGE	JUNCTION BLOCK AND WIRE HARNESS (CONNECTOR LOCATION)
1A	20	COWL WIRE AND J/B NO. 1 (LEFT KICK PANEL)
1E		
1H	20	ENGINE ROOM MAIN WIRE AND J/B NO. 1 (LEFT KICK PANEL)
2A	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2C	22 (2WD)	ENGINE WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2D	22 (2WD)	ENGINE ROOM MAIN WIRE AND J/B NO. 2 (NEAR THE BATTERY)
2E		
3A	25	COWL WIRE AND J/B NO. 3 (BEHIND COMBINATION METER)

□ : CONNECTOR JOINING WIRE HARNESS AND WIRE HARNESS

CODE	SEE PAGE	JOINING WIRE HARNESS AND WIRE HARNESS (CONNECTOR LOCATION)
EA1	36 (3S-GTE)	COWL WIRE AND ENGINE ROOM MAIN WIRE (FRONT SIDE OF RIGHT FRONT FENDER)
	38 (5S-FE)	
	40 (4A-FE)	
EB2	36 (3S-GTE)	ENGINE WIRE AND COWL WIRE (REAR SIDE OF RIGHT FRONT FENDER)
	38 (5S-FE)	
	40 (4A-FE)	
IG1	42	ENGINE WIRE AND COWL WIRE (UNDER THE ENGINE CONTROL MODULE (ENGINE ECU))
IH1	42	ENGINE WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IJ1	44	COWL WIRE AND A/C NO. 1 WIRE (BEHIND THE GLOVE BOX)
IQ2	44	COWL WIRE AND A/C NO. 2 WIRE (BESIDE HEATER UNIT)



: GROUND POINTS

CODE	SEE PAGE	GROUND POINTS LOCATION
EA	36 (3S-GTE)	FRONT RIGHT FENDER
	38 (5S-FE)	
	40 (4A-FE)	
EB	36 (3S-GTE)	FRONT LEFT FENDER
	38 (5S-FE)	
ID	42	LEFT KICK PANEL
IG	42	R/B NO. 4 SET BOLT



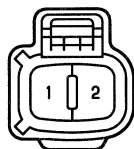
: SPLICE POINTS

CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS	CODE	SEE PAGE	WIRE HARNESS WITH SPLICE POINTS
E 2	36 (3S-GTE)	ENGINE ROOM MAIN WIRE	E 7	38 (5S-FE)	ENGINE ROOM MAIN WIRE
	38 (5S-FE)		E10	36 (3S-GTE)	COWL WIRE
	40 (4A-FE)			38 (5S-FE)	
E 3	36 (3S-GTE)			40 (4A-FE)	
	38 (5S-FE)		E13	36 (3S-GTE)	ENGINE ROOM MAIN WIRE
	40 (4A-FE)			38 (5S-FE)	
E 4	36 (3S-GTE)			40 (4A-FE)	
	38 (5S-FE)		I 3	44	COWL WIRE
E 5	36 (3S-GTE)	ENGINE WIRE	I 8		
	38 (5S-FE)		I 9		
	40 (4A-FE)		I12	44	A/C NO. 1 WIRE

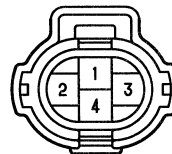
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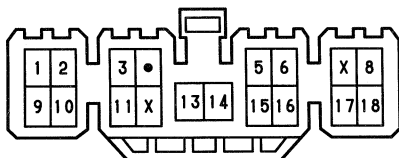
A 3 BLACK (5S-FE) A 4 (A) GRAY (4A-FE) A 4 (B) GRAY



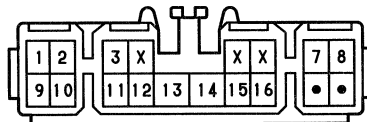
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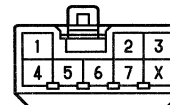
A15



A16 (A)



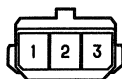
A18 (B)



A24



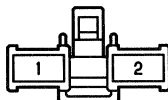
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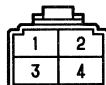
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B 4 (A) BLACK



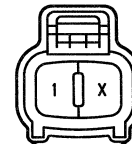
B 6 BLACK



C 1 (A) GRAY



C 2 (B) GRAY

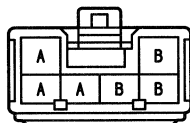


J 4



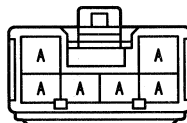
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J 5



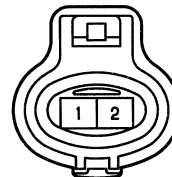
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J 6



(HINT:SEE PAGE 7)

R 1 GRAY



(EX. 4A-FE) W 3 DARKGRAY (4A-FE) W 3 GRAY

